# Green Home Score & Retrofit Recommendation Engine

*Mid-evaluation 2: real-data pipeline built end to end, energy models trained and honestly scored on held-out metered consumption*

## 1. Current Approach and Changes Since M1

The architecture from M1 stands: a deterministic engine produces one PropertyReport per home, and the LLM only explains it. What is now built on **real public data**:

- **Property book:** 542,711 homes from energy performance certificates in three contrasting local authorities, free-text fields mapped to model inputs (99.3-100% coverage), deprivation index joined by postcode.
- **Sales and values:** 175,457 house sales matched to homes (92.2% match rate; property type agrees for 93%), using only the certificate lodged *before* each sale. Current value for 153,511 homes = last sale x a price index for that property type and area.
- **Observed retrofit labels:** 4,791 real loft, cavity-wall and solar installs with metered use 2 years before vs after, kept in the fixed train/val/test split. Synthetic data only where none exists publicly (monthly bills, customer IDs, smart-thermostat level, loan-to-value), never used to report accuracy.

### Changes since M1, and why

- **Physics added back as V1 (grey box).** Heat loss per element (roof, walls, windows, floor, ventilation) x regional degree days, times a learned residual. V2 is the tuned gradient-boosted tree. Both are scored on the same folds; V1 ships because it models every retrofit physically and drives the heat-loss house diagram.
- **Two prediction modes.** *Similar homes* (description only) for homes without bills and for the 'worse than similar homes' detector; *this home* adds the home's own last-year bill, for existing customers.
- **Fairer consumption sub-score.** Raw kWh/m$^2$ penalised small homes: on real metered use the smallest homes scored 23.5 points below the largest. It is now actual / expected for similar homes: gap 6.6.
- **Feature parity and weather.** One feature definition for training and deployment (tested to match). Each home's gas bills are weather-normalised by a degree-day regression, as the national metered data is.
- **Unbiased totals.** Log-scale models predict the typical home, which undercounted portfolio totals by 12-17%; a smearing correction fixes it. Engine fixes: heat-pump savings in electrically heated homes, savings taken off the fuel the heating uses, and what-if scenarios now respect feasibility rules.

## 2. Experiments and Results

Training data: 50,000-home national sample with metered 2023 gas and electricity; fixed 70/15/15 split; 5-fold CV on train for every choice; test split not yet touched.

| Model (validation split, real metered use) | Gas MAE (kWh) | Gas $R^2$ (log) | Gas within 20% | Elec MAE (kWh) | Elec $R^2$ (log) | Elec within 20% |
|---|---|---|---|---|---|---|
| **Global median (no model)** | 4,930 | -0.01 | 27.6% | 1,714 | 0.00 | 25.0% |
| **Median of similar homes (no model)** | 4,131 | 0.26 | 33.4% | 1,614 | 0.11 | 27.1% |
| **Log-linear** | 4,029 | 0.30 | 33.7% | 1,618 | 0.11 | 27.0% |
| **Physics x one scale factor** | 4,439 | 0.16 | 31.8% | 1,653 | 0.03 | 25.7% |
| **V2: LightGBM, Optuna-tuned** | 3,972 | 0.30 | 35.1% | 1,580 | 0.16 | 27.6% |
| **V1: physics x tuned residual** | **3,967** | **0.30** | **34.9%** | **1,578** | **0.16** | **27.6%** |
| **This home: last year's bill (no model)** | 1,974 | 0.68 | 70.7% | 969 | 0.53 | 58.5% |
| **This home: change model (GBDT, L1)** | **1,923** | **0.71** | **71.2%** | **942** | **0.57** | **59.0%** |

- **The inputs cap description-only accuracy.** Homes with identical type, age, size, rating and region still vary so much that these bands explain at most $R^2$ 0.25 (gas) / 0.12 (electricity); V1 reaches 0.30 / 0.16. Optuna (50 trials per fuel) gained 1.1% on gas and nothing on electricity. Occupants, not buildings, drive the rest, so we report the ceiling rather than overclaim.

- **The bill is the strongest predictor.** 'Last year's bill' alone gets 70.7% of gas homes within 20%; plain models with the bill as a feature lost to it, so we predict the year-on-year change instead, which beats it by 2.6% (CV and validation).
- **Physics finding (prebound effect):** steady-state physics overpredicts metered gas about 2x (fitted scale 0.51), as documented for real homes heated less than assumed. Retrofit check on 2,968 observed installs: loft top-up observed -3.1%, V1 predicts -3.3%, V2 cannot (no loft input); cavity wall observed -10.9%, V1 -4.7%, physics alone -28%.
- **Portfolio level:** single-home errors cancel: total gas for 100 homes within 3.1%, for 1,000 homes within 0.9%. Picking the top 25% of users: precision 0.55, recall 0.54 for gas (0.47 / 0.46 electricity) vs 0.25 random.

## 3. Testing Progress

- **128 automated tests pass:** score weights and monotone mappings, size-neutral consumption score, feasibility rules, carbon, annualisation, a hand-checked heat-loss calculation, physics retrofit directions, training/deployment feature parity, degree-day fit recovery, schema and API.
- **Data verification:** 29 checks recomputed independently from the raw files (match rate, no sale uses a later certificate, 2,000 valuations recomputed exactly, loan balances, label splits); a clean rebuild reproduces every committed summary. 3,000 real homes run through the engine with 0 errors; per-retrofit savings sum to the package total for every home.
- **Review passes found and fixed real bugs:** future months parsed as fake degree days, 282 gas-heated homes with no gas meter, heat pumps raising electricity in electrically heated homes, an evaluation script that silently skipped electricity. Known open items are tracked, e.g. the grounding checker's sign handling.

## 4. Plan for the Remaining Time

- **Energy model:** conformal p10/p90 bands (target ~80% coverage); one model interface wired into the savings simulator; retrofit check on held-out installs; then a **single** test-set scoring and the held-out predictions file, with errors by region, property type, age and rating for the fairness audit.
- **Value model:** quantile gradient-boosted price level plus a hedonic rating-band uplift, trained on the 168,351 sales with a pre-sale certificate; interval coverage vs a regional-median baseline.
- **Detector and engine:** 'worse than similar homes' from actual / expected with a band-based threshold; remove placeholder figures from the dashboard; fix grounding sign handling and per-home memory.
- **LLM and delivery:** tool-calling agent over the report JSON with the grounding gate, the ~50-scenario LLM evaluation, fairness write-up, README, final report and demo.

**Reproducibility.** *make build* rebuilds the whole real data layer (~2.5 min), *make train* trains and scores every model, *make test* runs the suite. Splits are fixed and versioned, and every constant in the configuration carries a source.